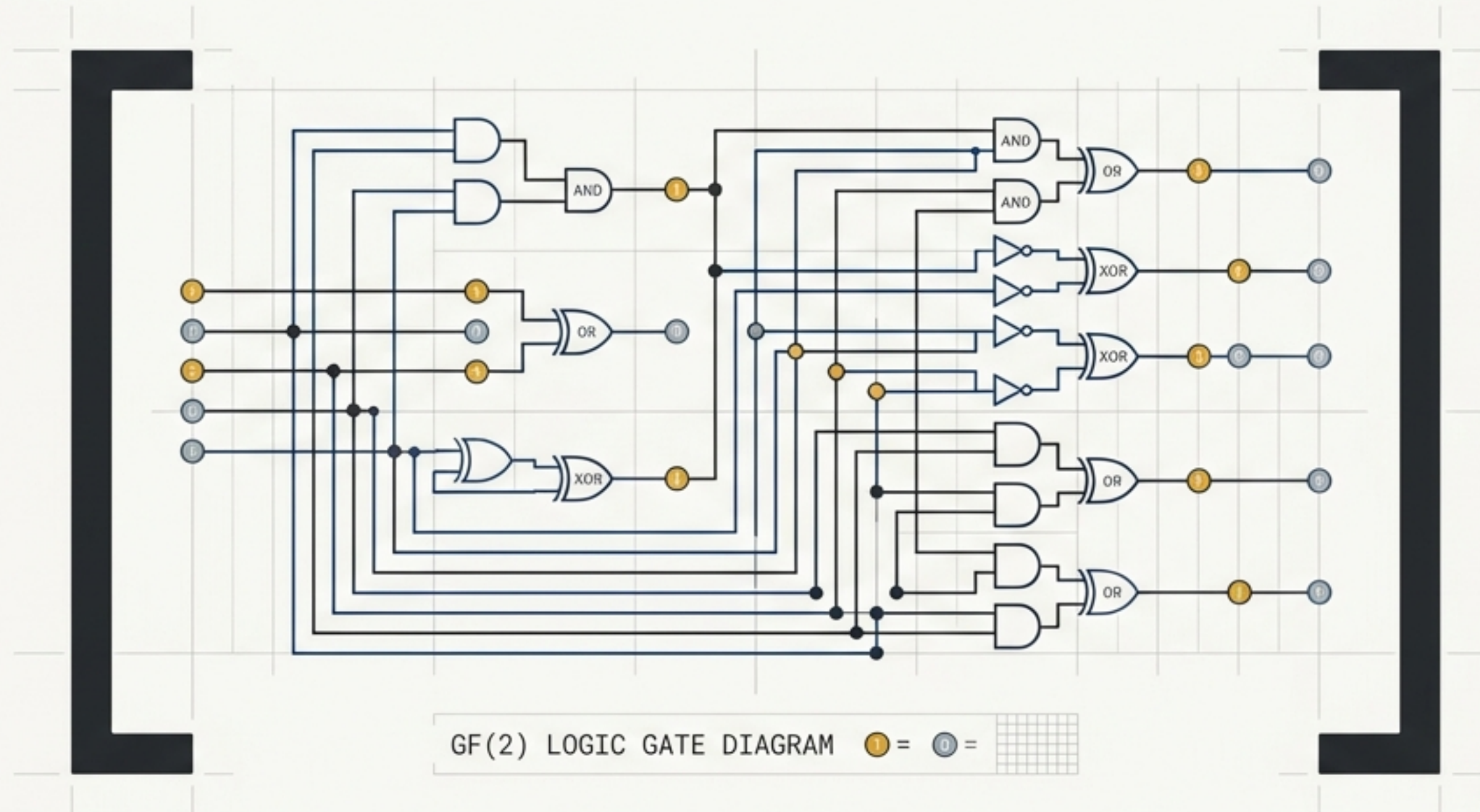


Linear Transformations & Matrices over GF(2)

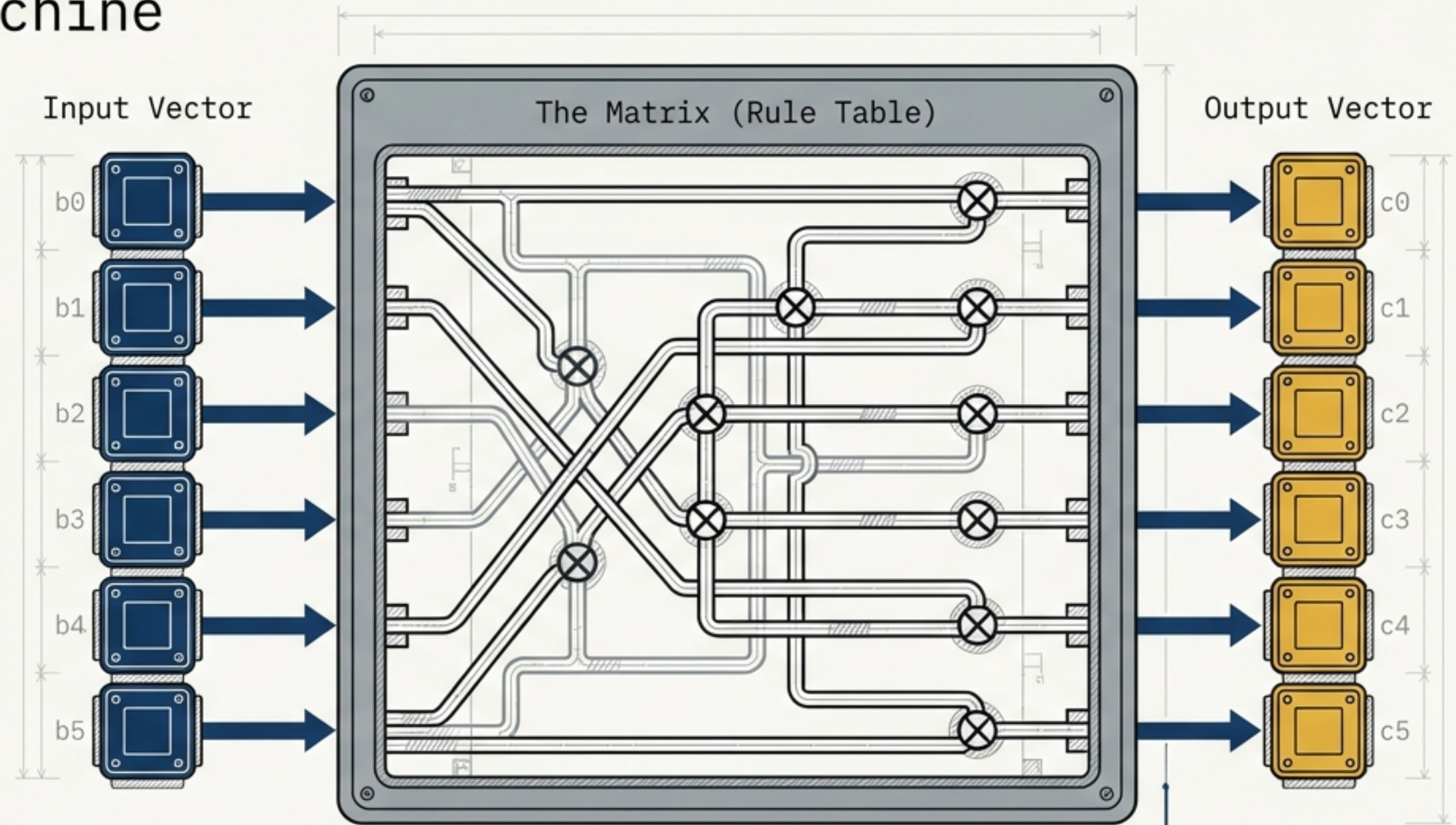
The Disciplined Wiring of Cryptography



The Bit-Mixing Machine

AES and other ciphers rely on structured bit mixing.

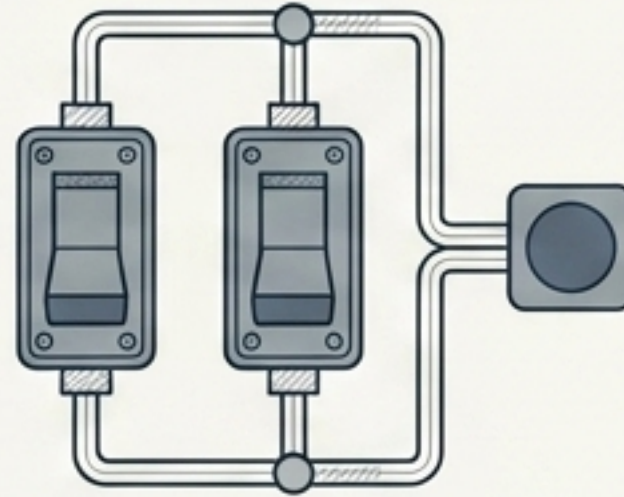
A matrix over $GF(2)$ is simply a rule table made of bits. It acts as a machine that takes input bits, combines selected ones with XOR, and produces output bits.



Zero carrying. Zero decimal math.
Just structured XOR wiring with algebraic manners.

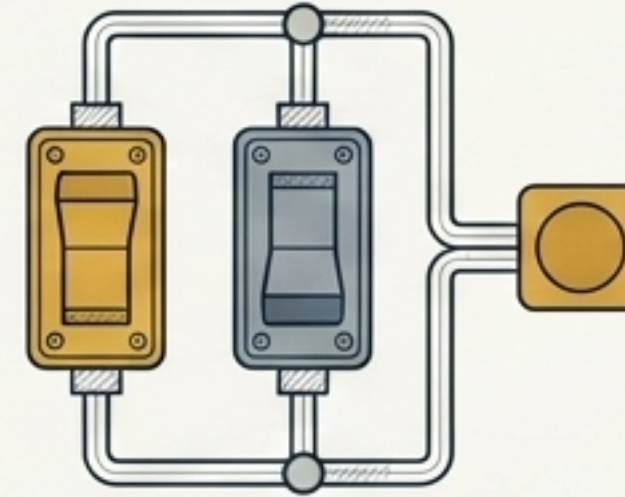
The Physics of GF(2): Modulo 2 Arithmetic

In the finite field of two elements (GF(2)), the only numbers are 0 and 1, and addition is entirely replaced by XOR ($\oplus$).



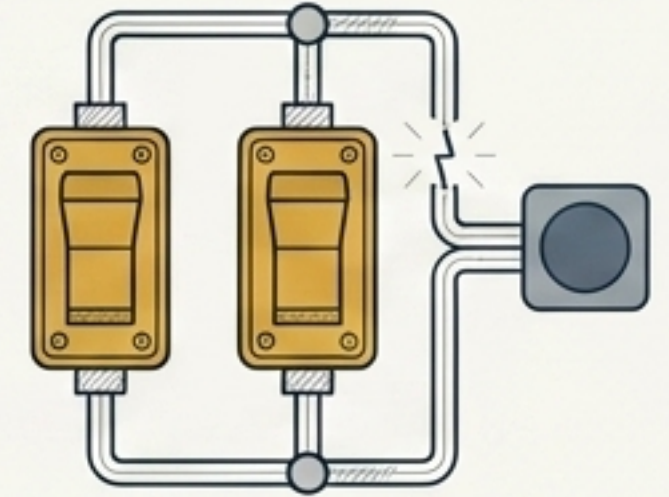
$$0 + 0 = 0$$

(Two open switches = Off)



$$1 + 0 = 1$$

(One closed switch = On)



$$1 + 1 = 0$$

(Two closed switches trigger a breaker, canceling out = Off)

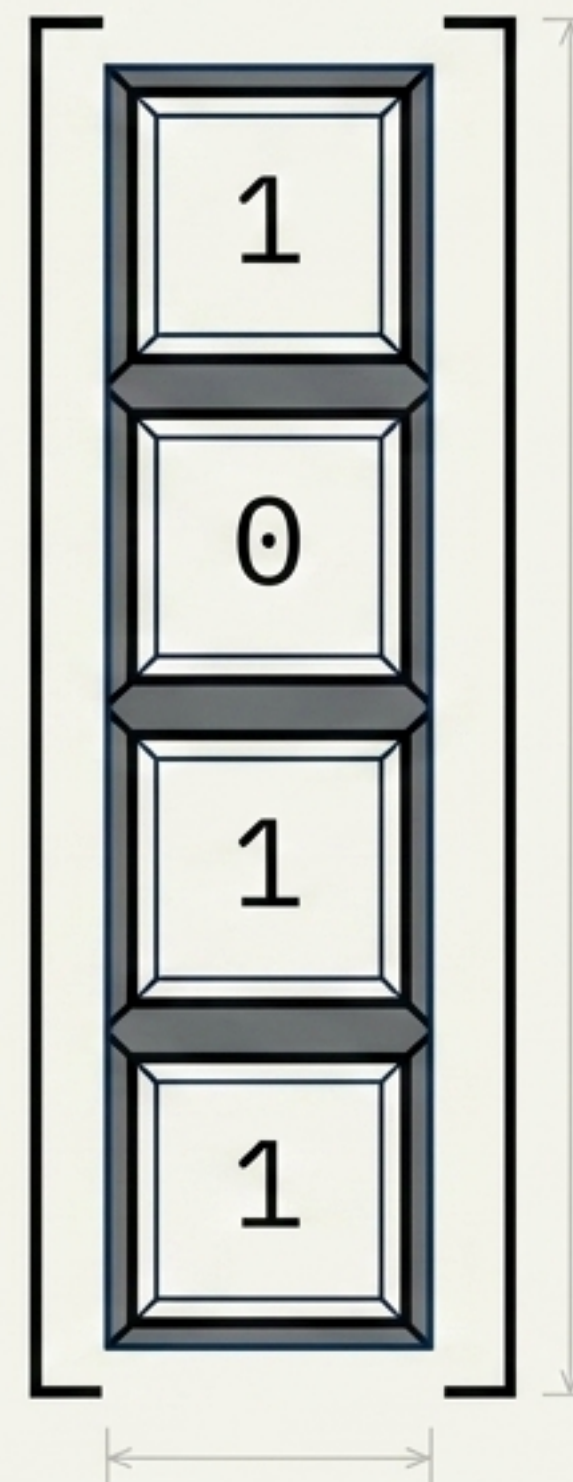
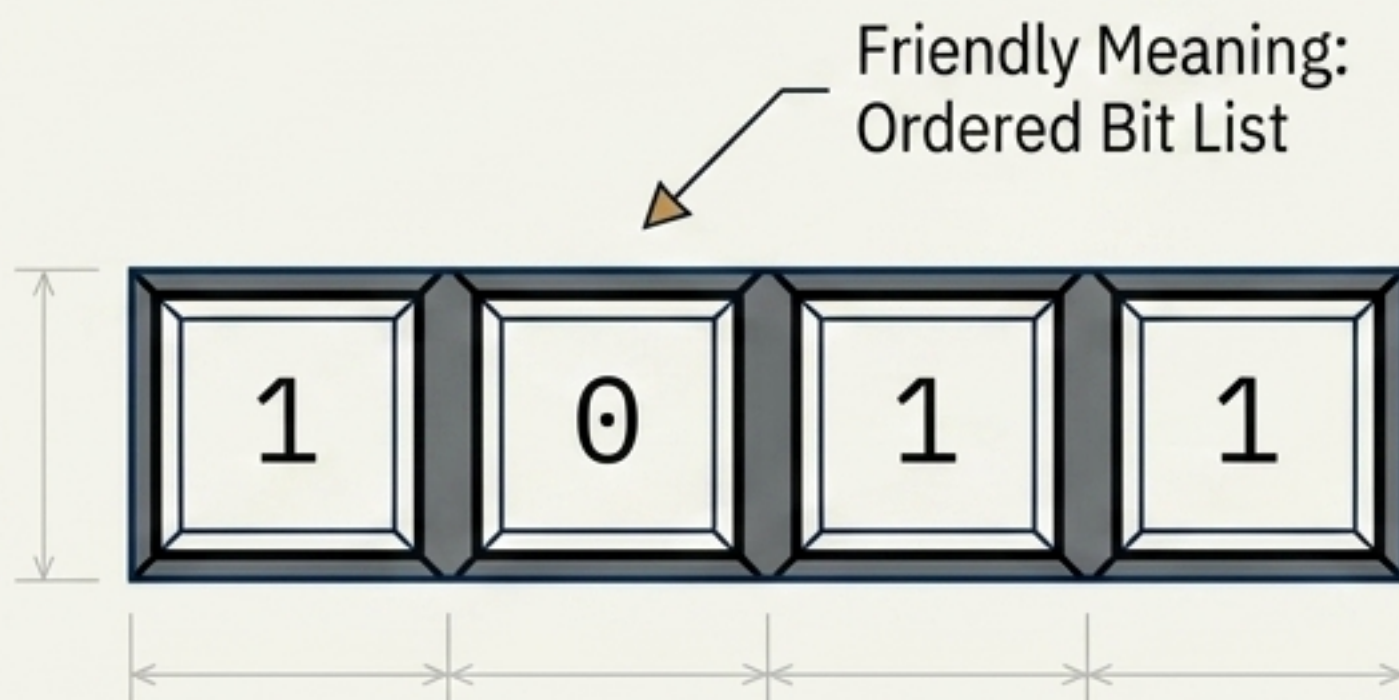
The Shortcut:

- XORing a group of bits yields 1 if an odd number of bits are active, and 0 if an even number are active.

$$1 \oplus 1 \oplus 0 \oplus 1 = 1$$

The Raw Material: Vectors as Bit Stacks

A vector is simply an ordered list of bits. We stack bit vectors vertically because this orientation aligns perfectly with the intake mechanisms of matrix multiplication.



Technical meaning:
4-dimensional
vector over GF(2)

$$v = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

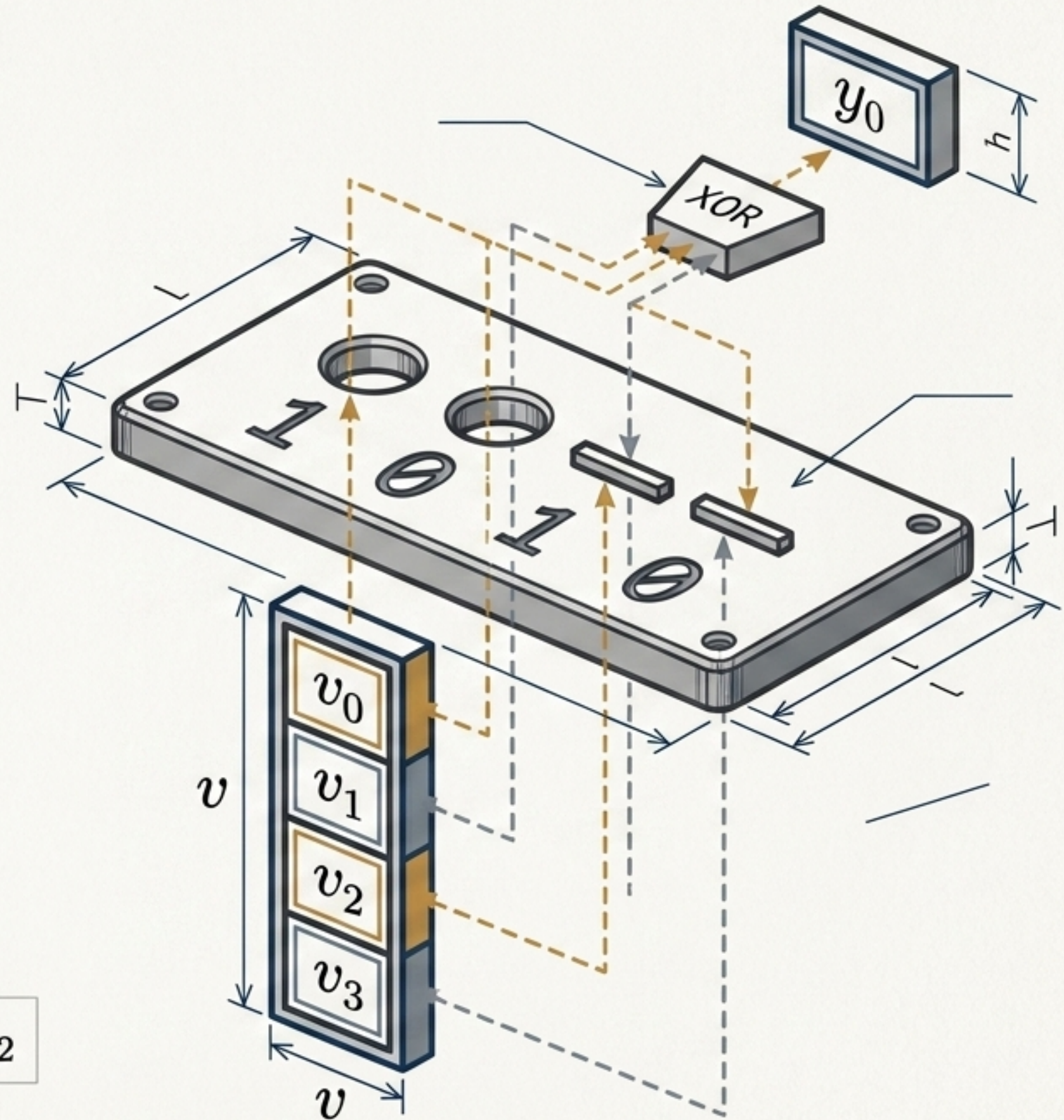
The Mechanism: Every Matrix Row is a Recipe

Each row of a matrix generates exactly one output bit. The binary rule is absolute:

- **1** = Include the corresponding input bit.
- **0** = **Ignore** the corresponding input bit.

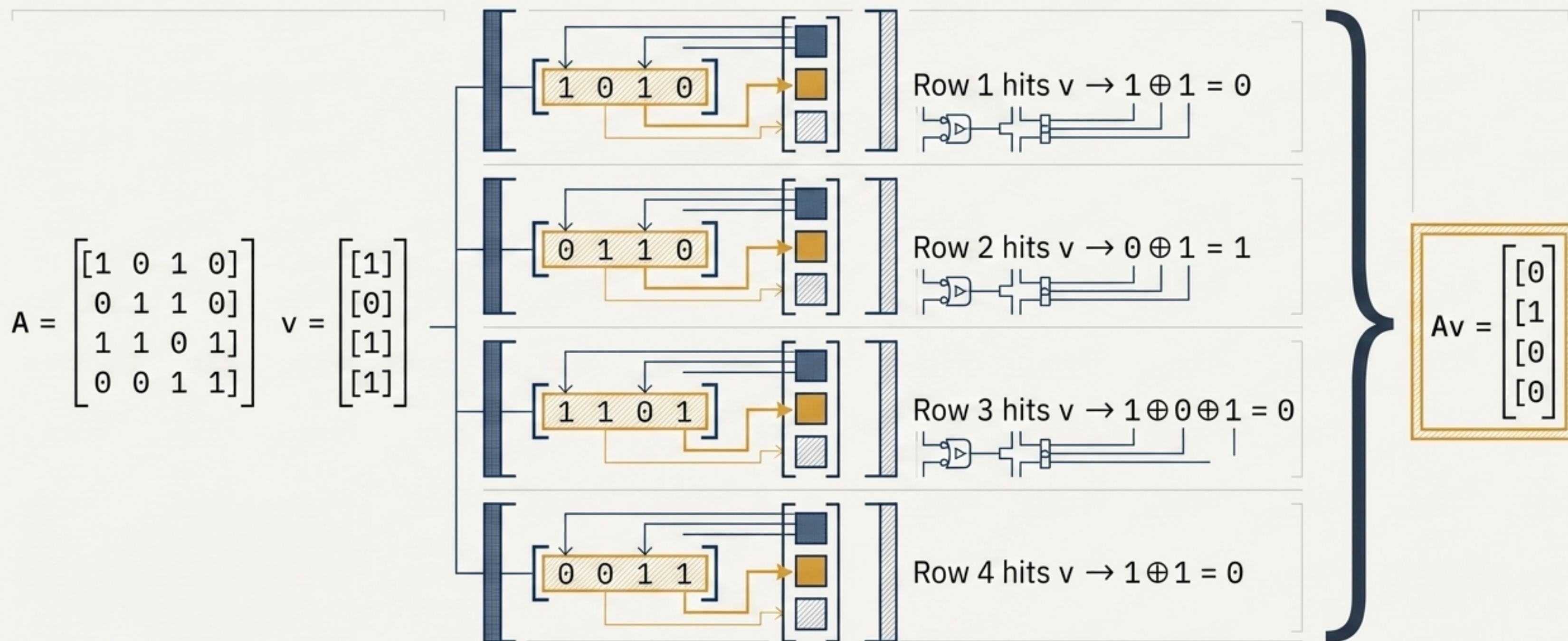
$$y_0 = 1v_0 \oplus 0v_1 \oplus 1v_2 \oplus 0v_3$$

$$y_0 = 1v_0 \oplus 0v_1 \oplus 1v_2 \oplus 0v_3 \rightarrow y_0 = v_0 \oplus v_2$$



Executing the Blueprint: Matrix-Vector Multiplication

Running the full matrix applies every row's recipe to the input vector simultaneously, generating a completely new state.



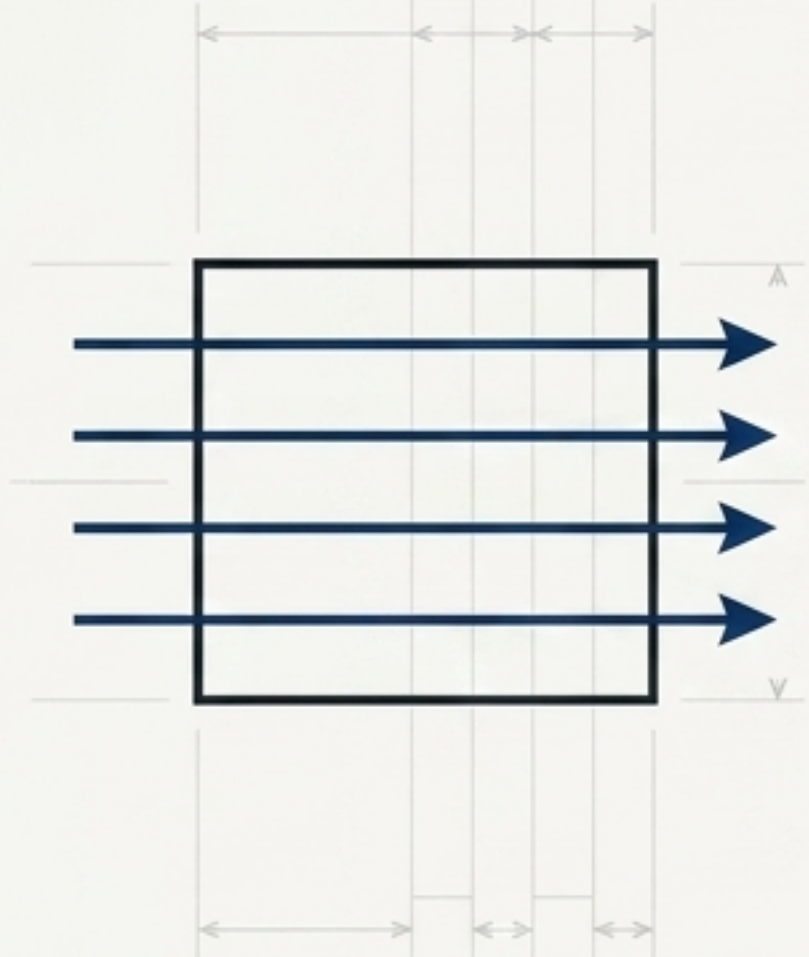
Archetype I: The Identity Matrix

The “Do Nothing” machine. It keeps each bit in its exact original position.

- Every row selects exactly one input bit and sends it straight through. It acts as a perfect wire passthrough.

For any vector $\mathbf{v}$,
 $\mathbf{I}\mathbf{v} = \mathbf{v}$.

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

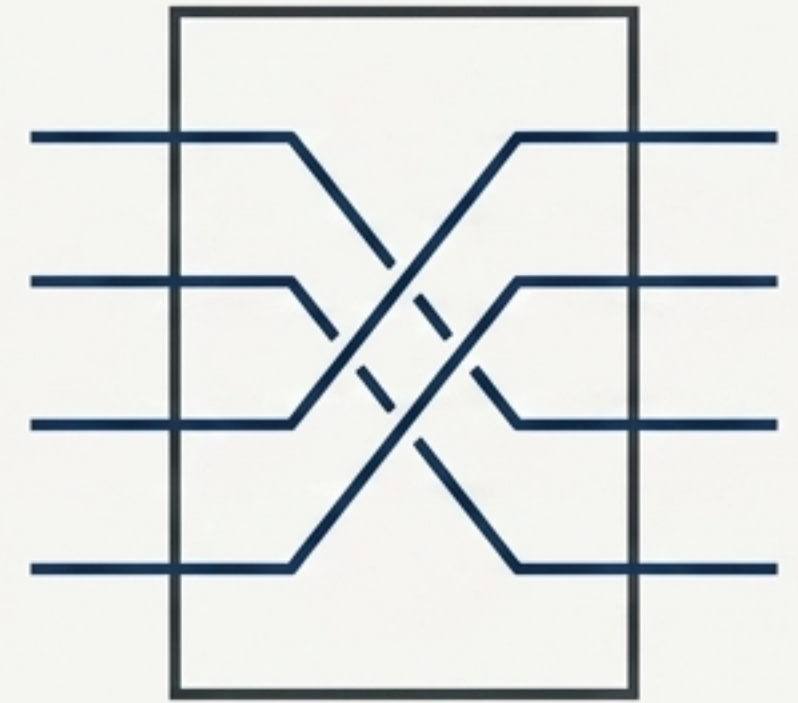


Archetype II: The Permutation Matrix

The 'Rearranger'. It scrambles bit positions without mixing them together.

A permutation matrix always has exactly one 1 in each row and column. It is strictly structural routing and is always 100% reversible.

$$P = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

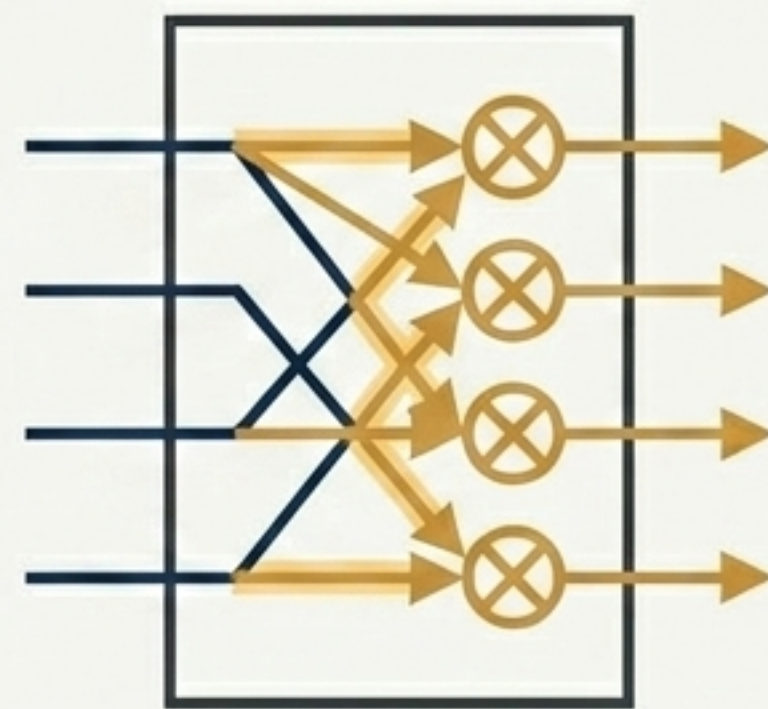


Archetype III: The Mixing Matrix

The foundation of cryptographic diffusion. These matrices do more than rearrange—they force bits to collide.

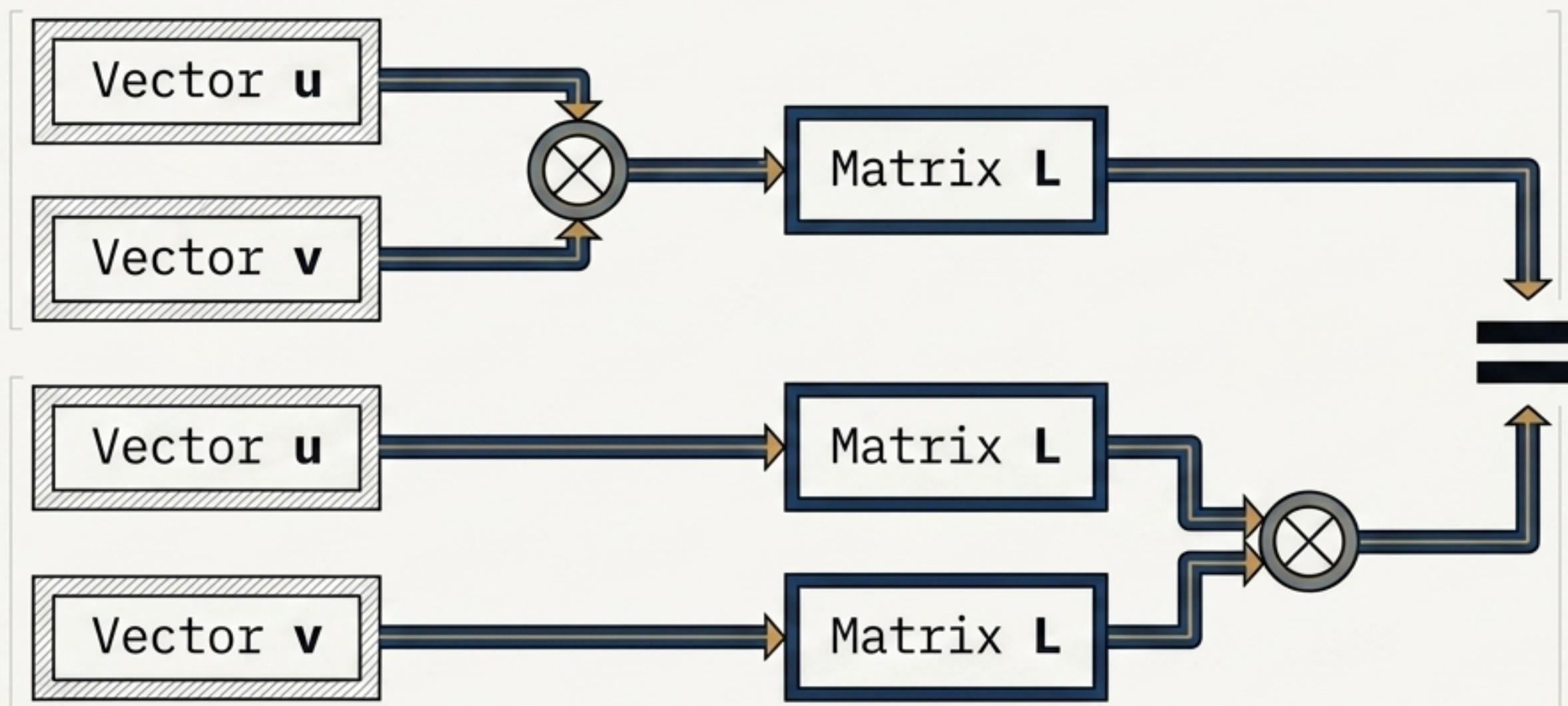
- Because each output row depends on multiple inputs (e.g., $y_0 = v_0 \oplus v_1$), a single bit flip in the input will avalanche across multiple outputs.

$$M = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$



The Rule of Linearity

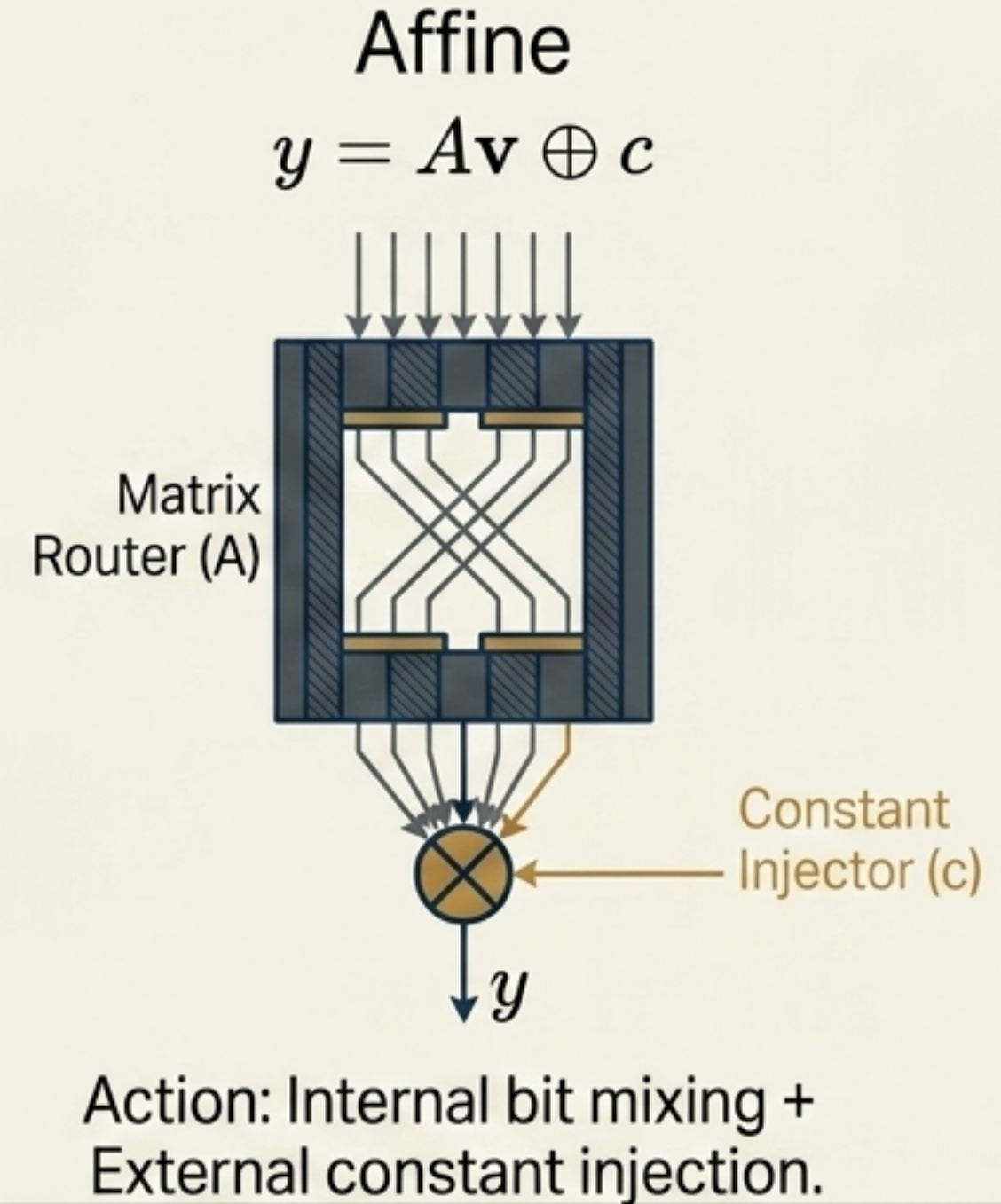
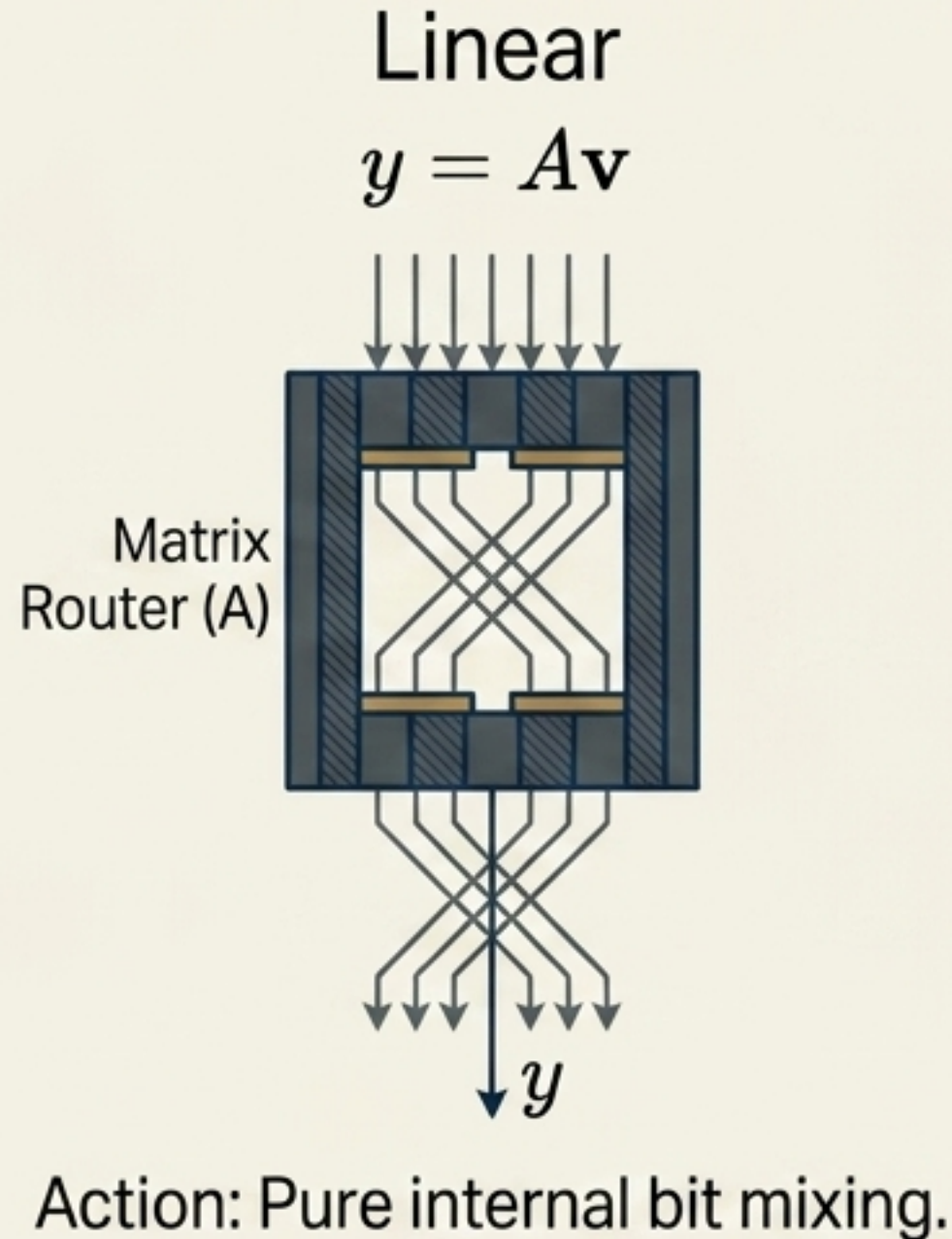
A transformation is strictly linear if it respects XOR addition. If you XOR inputs first and then transform, you get the exact same result as transforming each input separately and then XORing the outputs.



$$L(\mathbf{u} \oplus \mathbf{v}) = L(\mathbf{u}) \oplus L(\mathbf{v})$$

System Upgrade: Linear vs. Affine Transformations

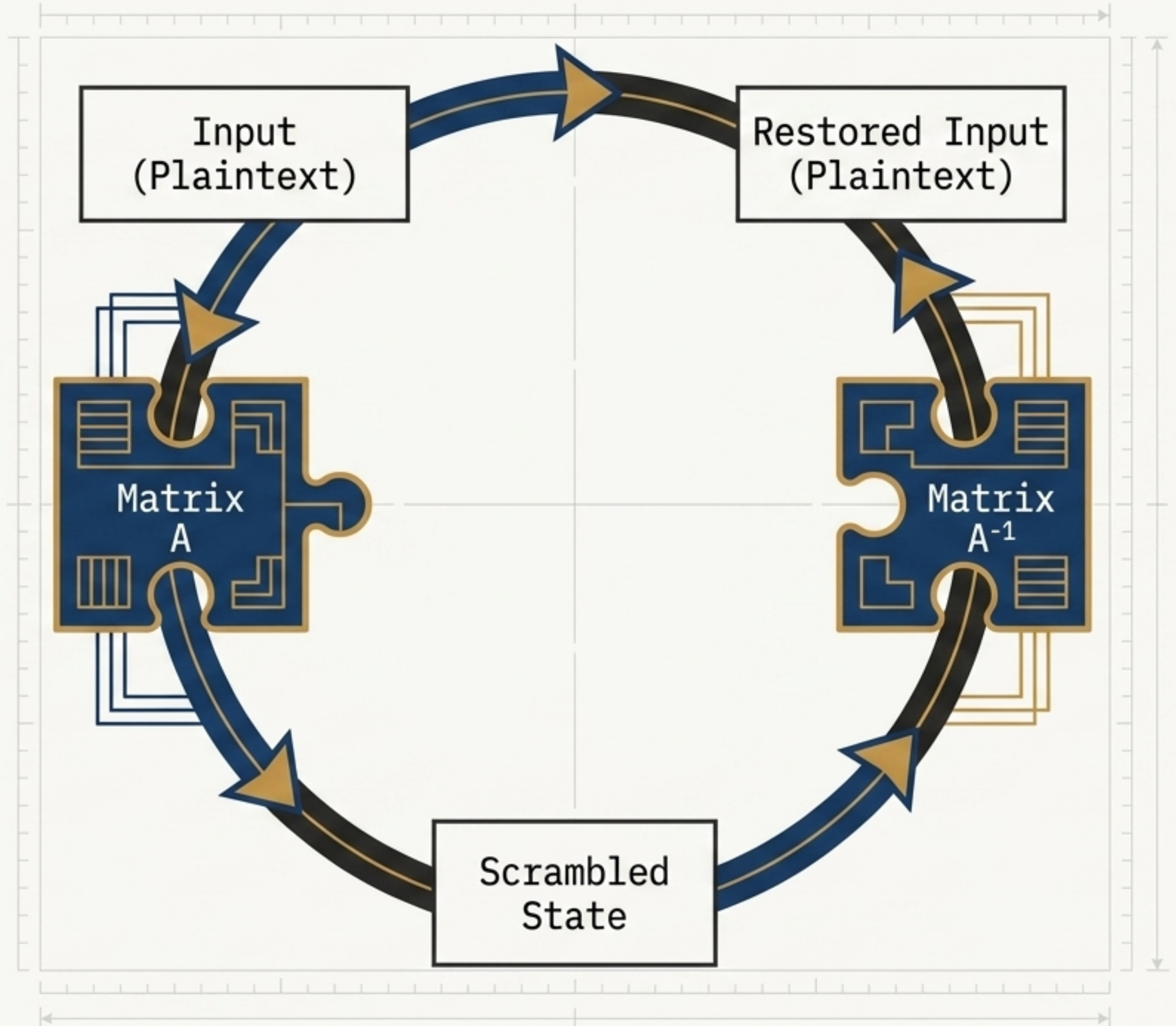
The AES S-box operates precisely on this upgrade. It uses a nonlinear inverse step, followed by an affine transformation (a matrix-based bit mix, plus the fixed constant 0x63).



The Undo Button: Invertible Matrices

If a cipher collapses information, data is lost forever. For a block cipher to be decrypted, its mixing layers must be entirely invertible.

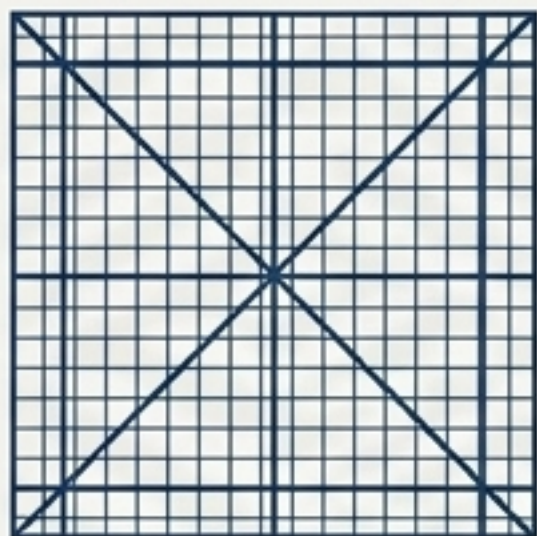
- If Matrix A mixes the bits, there must exist an Inverse Matrix A^{-1} that perfectly undoes the wiring: $A^{-1}A = I$.



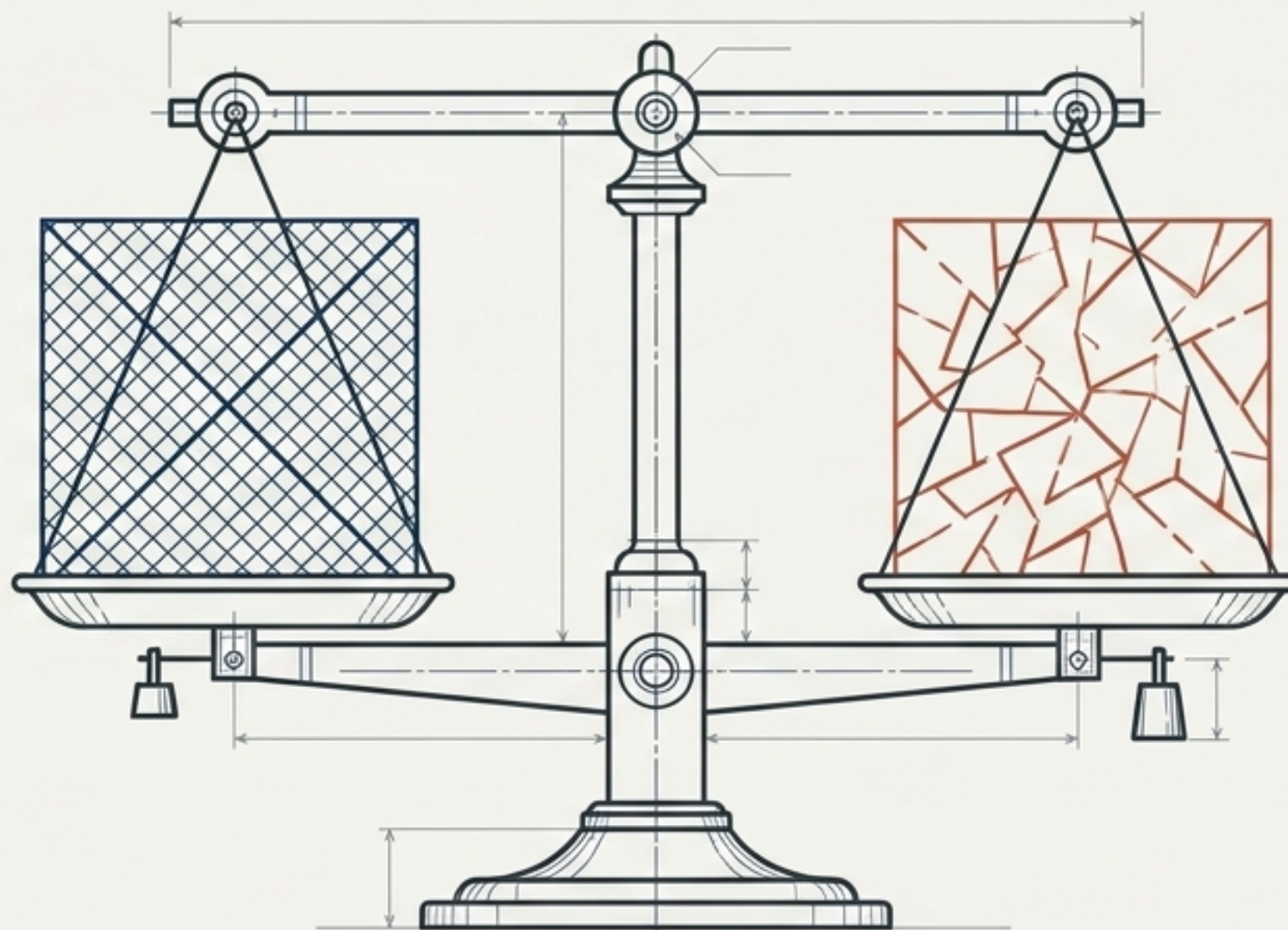
The Weakness of Purity

A common cryptographic mistake is assuming "linear" means "weak." Linear layers are incredibly powerful for diffusion—but if a cipher relies on linear operations alone, it can be trivially solved with basic algebra. Strong ciphers require an alliance of structured linear mixing and non-linear disruption.

Linear Layers



Predictable.
Spreads influence
(Diffusion).



Non-Linear Layers

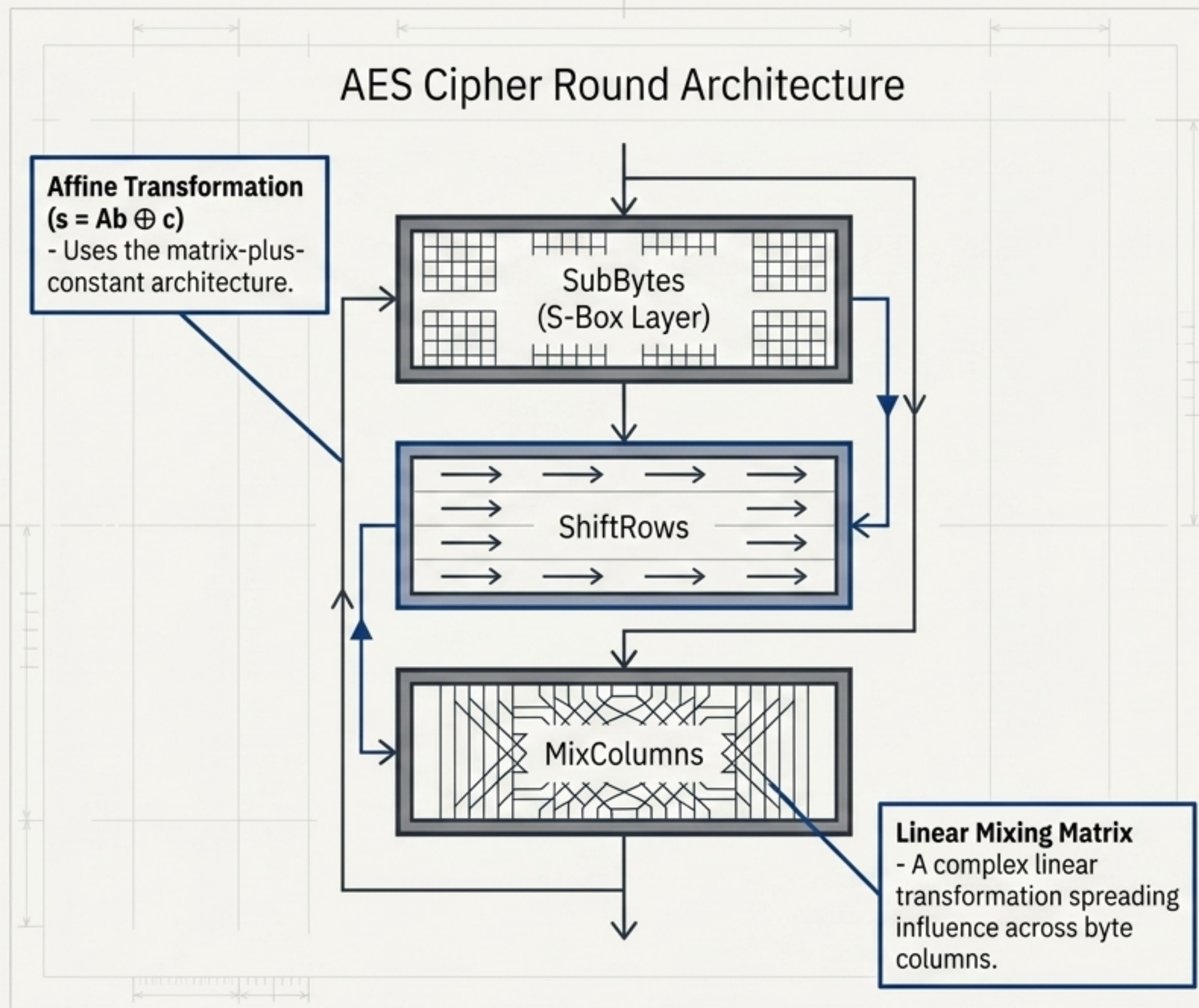


Unpredictable disruptors.
Destroys relationships
(Confusion).

Real-World Architecture: The AES Connection

Matrix mechanics aren't abstract theory—they are the literal blueprints of the Advanced Encryption Standard.

AES utilizes invertible affine transformations, invertible S-boxes, and invertible linear diffusion matrices to secure data.



The Cryptographic Blueprint: Reference Manual

Concept	Plain English	Crypto Connection	Core Formula
Vector	Bit List	State bits	$v = [1, 0, 1]^T$
Matrix	Rule Table	Structured bit mixing	A
Linear	Pure Mixing	Diffusion Layers	$y = Av$
Affine	Mix + Constant	AES S-Box layer	$y = Av \oplus c$
Invertible	Undo Button	Decryption requirement	$A^{-1}A = I$

Matrices provide the concrete mechanism for spreading bits. Next, we measure the quality of that spread in: Diffusion, Branch Number, and Avalanche.